

ASTRO 1: The Astronomical Universe

Instructor: Chloe Klare

Fall 2025

Course Description

This course meets the Bachelor of Arts general education requirement for non-science majors. We will cover an introduction to a broad range of topics in astronomy and astrophysics, emphasizing *how* we have learned about our universe. Planned topics include: Motions of the Earth, and how this affects our view of the universe; light, the electromagnetic spectrum, and telescopes; the Solar system and its constituents; anatomies and life cycles of stars; galaxies; the early universe and the expanding universe; and multi-messenger astronomy.

Learning Objectives

At the end of this course, students will be able to...

- ⊙ understand and explain various introductory topics in astronomy and astrophysics
- ⊙ conceptually reason through and quantitatively solve astronomy problems
- ⊙ use their scientific literacy to critically approach ideas, claims, and theories in astronomy
- ⊙ apply the concepts covered in class to their own lives

This course fulfills the **Key Literacies** and **Critical and Analytical Thinking** general education learning objectives; specifics on these can be found under the [Undergraduate Bulletin](#).

Course Logistics

Instructor Information

Instructor:

- ⊙ Chloe Klare (she/her)
- ⊙ Contact:
- ⊙ Office Location:
- ⊙ Office Hours: Monday 11am-12pm, Tuesday 5:30pm-6:30pm, or by appointment

Additional office hours

The astronomy department offers general astronomy office hours for several hours per week, located in 443 Davey Lab. All general TAs should be able to help with any material from this course, but note that they are not TAs specific to this course, so please bring any material/homework with you. Exact times are listed on Canvas.

Course Meetings

- ⊙ Meeting Times:
- ⊙ Meeting Place:

Course Materials

There are no required course materials. *Astronomy* by Fraknoi, A. *et al.* is a helpful open source online textbook for optional supplemental reading. A standard calculator will be helpful for homework assignments and exams.

Course Assessments

In-Class Assignments (10%)

These will be closed-book, closed-note, 15-minute pop-quiz type written assignments given at the end of class on a ~weekly basis. They will generally consist of 1-2 questions asking students to explain topics recently covered in the course. They will be **graded on effort**, and they serve the following purposes:

- ⊙ Provide an opportunity for students to engage with the material they are currently learning
- ⊙ Enable the instructor to gauge students' progress towards the course learning objectives, and adjust as necessary
- ⊙ Allow the instructor to give constructive feedback in a low-stakes manner
- ⊙ Prepare students for the exams, which will ask some of the same types of questions

In-class assignments must be submitted on paper prior to leaving class, or they will not be accepted.

Homework Assignments (15%)

Homework will be assigned biweekly, and will consist of a mixture of quantitative and conceptual problem solving. Students may collaborate with others, but each student is expected to submit their own original work. Solutions without justification will receive no credit. Note that submitting unoriginal work (including work done using prohibited uses of AI) will result in academic integrity sanctions and inadequate preparation for the exams. **Homework questions are very likely to be featured later on the exams.** Assignments should be submitted as a PDF via Canvas. These assignments serve the following purposes:

- ⊙ Reinforce students' understanding of course material
- ⊙ Challenge students with more complex questions than in-class assignments
- ⊙ Allow the instructor to give constructive feedback and identify concepts students may be struggling with
- ⊙ Prepare students for the exams

Exams (50% total)

There will be three exams - two midterms and a cumulative final - and they will be closed-book, closed-note. At the instructor's discretion, they may be curved and/or opportunities to earn back points will be provided. Midterm exams will be in-class on **October 3** and **November 14**. The final exam is TBD during the week of December 15. Students may bring an equation sheet (1 page) and a calculator.

Final Project (15%)

The final project will consist of a 2500 word synthesis paper on a subtopic in astronomy. More details and topic ideas are available in Canvas. The goal of this project is for students to engage in-depth with a topic of interest to them.

Astronomy in the Wild (10%)

Students will complete an activity of their choosing incorporating astronomy into their daily life outside of the classroom and reflect on how it relates to the course material. A full list of activity options and requirements are listed in Canvas.

Extra Credit

The following extra credit opportunities will allow students to earn up to 0.5% extra credit towards their overall course grade. More opportunities may be offered throughout the semester at the instructor's discretion.

- ⊙ Submit an additional Astronomy in the Wild assignment
- ⊙ Attend the mid-semester STAR party (observing on the roof of Davey Lab) on Saturday, November 15
- ⊙ If > 90% of the class completes the end of semester feedback form (SEEQ), everyone will receive 0.5% extra credit towards their final grade

Grading Policies

Grade Distribution

Category	Weight
Midterm exams (2)	30%
Final Exam (cumulative)	20%
Homework assignments	15%
In-class assignments	10%
Final project	15%
Astronomy in the Wild	10%+
Total	100%+

Grading Scheme

Grade	Percent Range
A	$\geq 93\%$
A-	$\geq 90\%$
B+	$\geq 88\%$
B	$\geq 83\%$
B-	$\geq 80\%$
C+	$\geq 78\%$
C	$\geq 70\%$
D	$\geq 60\%$
F	$< 60\%$

Course Policies

Attendance Policy

Attendance will not be recorded, but note that **attending lecture is the best practice for successful learning** in this course. Additionally, in-class assignments cannot be made up due to absences. However, points can be earned back for missed in-class assignments via extra credit opportunities, so students should not fret over occasional absences due to travel or illness (and should please not attend class if sick!).

Late Work Policy

Late homework assignments **only** will be accepted within 1 sidereal day (23 hours, 56 minutes) for a maximum of half-credit. No late homework assignments will be accepted after this period, and no assignments in any other category will be accepted after the deadline.

Academic Integrity Policy

Every student who is trying in this course should be able to pass, and demonstrating the correct thought processes and logical reasoning is almost always more important than having the right answer. There is no excuse for cheating, plagiarism, or submitting assignments produced by ChatGPT in this course. The first violation will result in a 0 on the offending assignment, and any additional violations will result in an F in the course. The full Penn State policy on academic integrity can be found [here](#); note that students with open academic misconduct cases may not withdraw from the affected course.

Curving Policy

Final grades will not be curved. Exam grades may be curved if the distribution is centered around $\lesssim 75\%$.

Technology + Generative AI Policy

The use of cell phones, laptops, or other technology during exams and in-class quizzes is prohibited. Scientific and/or graphing calculators only are allowed on exams. The use of generative AI tools is not allowed on course assignments, and violations will be treated as academic integrity violations. Language models, such as ChatGPT, are designed to predict the most likely output for a given input based on their training set. They do not, and cannot, provide new information, and are not designed to provide the most accurate information (like an internet search would). Note that as **exam questions will be similar to the homework questions**, using generative AI for prohibited purposes will leave students ill-equipped to succeed on the exams.

Academic Accommodations

Students with disabilities are welcome in all Penn State educational programs, including this course. For any student with academic accommodations through [Student Disability Services](#), the instructor will work with them to adapt course policies to ensure they can be successful in this class.

Resources for Student Support

Counseling and Psychological Services

[CAPS](#) services include (but are not limited to!) counseling, psychiatry, support groups, and crisis intervention, and they can be reached at 814-863-0395. The Penn State crisis line is open 24/7 and can be reached via phone at 877-229-6400 or text by texting LIONS to 741741.

Student Care and Advocacy

[Student Care and Advocacy](#) supports students in many situations, including food/housing insecurity, academic or health (both mental and physical) challenges, and more.

University Health Services

[University Health Services](#) provides a range of healthcare services, including mental health treatment, eating disorder/health eating support, and gender diverse care.

Health Promotion and Wellness

Penn State offers free wellness services (including nutrition, stress management, and substance abuse counseling) to all students through [Health Promotion and Wellness](#).

Community Resources

The Centre County [Center for Community Resources](#), located a short distance off-campus at 2100 East College Avenue, offers 24/7 walk-in (or phone-in) crisis services (and more!).

Educational Equity

Fostering a diverse and inclusive environment for students is a priority for both the instructor of this course and Penn State as a whole. Acts of intolerance, discrimination, or harassment due to age, ancestry, color, disability, gender, gender identity, national origin, race, religious belief, sexual orientation, or veteran status are not tolerated and can be reported through Educational Equity via the Report Bias [webpage](#).